RAGMate Academic Research Report: Dense Retrieval Architectures
Authors: MCA Research Laboratory

ABSTRACT
Retrieval-Augmented Generation (RAG) integrates dense passage retrieval with pre-trained sequence-to-sequer

1. INTRODUCTION
Standard large language models (LLMs) struggle with factual hallucinations and lack verifiable provenance wher

2. SYSTEM ARCHITECTURE & METHODOLOGY

The proposed architecture consists of four distinct processing tiers:
1. Document Ingestion: Arbitrary multi-page PDF documents are ingested and parsed using high-performance C
2. Page-level Text Normalization: Whitespace is regularized, hyphenated line breaks are resolved, and control ch
3. Recursive Chunking: Cleaned text is partitioned into overlapping windows of size 800 to 1200 characters with
4. Dense Semantic Embedding: Each chunk is encoded into a 384-dimensional vector space using the all-MiniLM

Vector indexing is performed in ChromaDB, enabling sub-10ms nearest neighbor search using cosine similarity m

3. EXPERIMENTAL EVALUATION

We evaluated retrieval efficacy across three paradigms: Keyword BM25, Dense Semantic Retrieval, and Dense

Results show that Cross-Encoder Reranking improves Mean Reciprocal Rank (MRR) from 0.74 to 0.89 while ma

4. CONCLUSION & FUTURE WORK

Page-grounded RAG provides an academically sound, hallucination-resistant architecture for document question